

AZURE-AWS S2S VPN

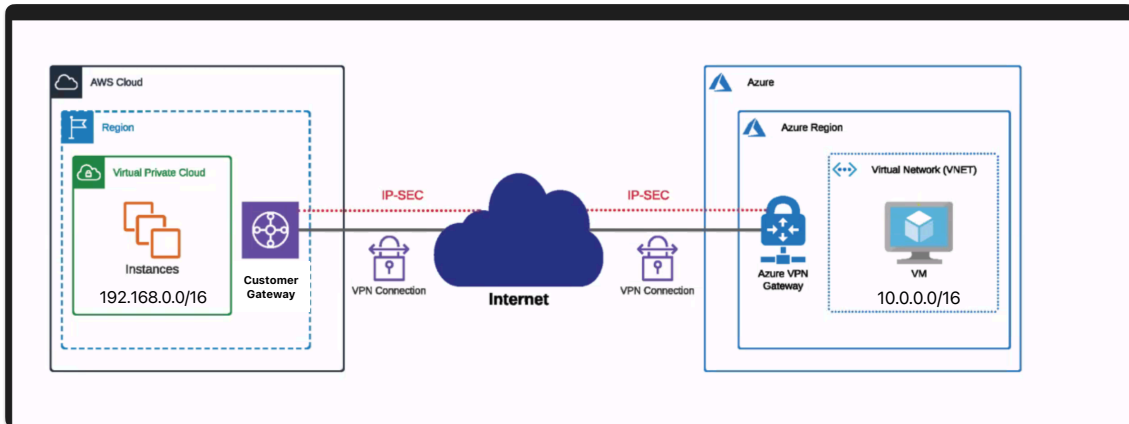
Batch: Azure 26-Nov

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Objective: To create Site-to-Site VPN from Azure VNET to AWS VPC.



Resources:

AZURE	AWS
❖ Resource Group ❖ Virtual Network 1. Private Subnet 2. GatewaySubnet ❖ VPN Gateway ❖ Local Gateway ❖ Connection ❖ VM Instance	❖ Virtual Private Cloud 1. Private Subnet 2. Internet Gateway 3. Routing Table ❖ VPN Gateway ❖ Customer Gateway ❖ S2S VPN Connection ❖ EC2 Instance

Steps to Create S2S VPN Connection:

Step 1: Create Resource Group RG-US on Azure

Microsoft Azure

Search resources, services, and docs (G+)

[Home](#) > [Resource groups](#) >

Create a resource group

Basics

Tags

Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Project details

Subscription * ⓘ

Azure subscription 1

Resource group * ⓘ

RG-US

Resource details

Region * ⓘ

(US) East US

Step 2: Create Virtual Network VNET-US on Azure under RG-US Resource Group

Microsoft Azure

Search resources, services, and docs (G+)

[Home](#) > [Virtual networks](#) >

Create virtual network

Basics

IP Addresses

Security

Tags

Review + create

The virtual network's address space, specified as one or more address prefixes in CIDR notation (e.g. 192.168.1.0/24).

IPv4 address space

10.0.0.0/16 10.0.0.0 - 10.0.255.255 (65536 addresses) 

☐ Add IPv6 address space ⓘ

The subnet's address range in CIDR notation (e.g. 192.168.1.0/24). It must be contained by the address space of the virtual network.

[+](#) Add subnet [Remove subnet](#)

☐ Subnet name	Subnet address range	NAT gateway
☐ subnet-US	10.0.0.0/24	-
☐ GatewaySubnet	10.0.1.0/24	-

i A NAT gateway is recommended for outbound internet access from subnets. Edit the subnet to add a NAT gateway. [Learn more](#)

Step 3: Create Virtual Network Gateway VPN-GW-US on Azure under RG-US Resource Group

Microsoft Azure

Search resources, services, and docs (G+/)

Home > Virtual network gateways >

Create virtual network gateway ...

Instance details

Name *

VP-GW-US

Region *

East US

Gateway type * ⓘ

☒ VPN ☐ ExpressRoute

VPN type * ⓘ

☒ Route-based ☐ Policy-based

SKU * ⓘ

VpnGw2AZ

Generation ⓘ

Generation2

Virtual network * ⓘ

VNET-US

Subnet ⓘ

GatewaySubnet (10.0.1.0/24)

Only virtual networks in the currently selected subscription and region are listed.

Public IP address

Public IP address * ⓘ

☒ Create new ☐ Use existing

Public IP address name *

vnetgw-pip

Public IP address SKU

Standard

Assignment

☐ Dynamic ☒ Static

Availability zone *

Zone-redundant

Enable active-active mode * ⓘ

☐ Enabled ☒ Disabled

Configure BGP * ⓘ

☐ Enabled ☒ Disabled

Azure recommends using a validated VPN device with your virtual network gateway. To view a list of validated devices and instructions for configuration, refer to Azure's [documentation](#) regarding validated VPN devices.

Step 4: Create Virtual Cloud VPC-AWS on AWS

VPC settings

Resources to create

☐ VPC only

☒ VPC and more

Name tag auto-generation

☒ Auto-generate

VNET-AWS

IPv4 CIDR block

192.168.0.0/16

65,536 IPs

IPv6 CIDR block

☒ No IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

Tenancy

Default

Number of Availability Zones (AZs)

1

2

3

Customize AZs

Number of public subnets

0

1

Number of private subnets

0

1

2

Customize subnets CIDR blocks

Public subnet CIDR block in ap-south-1a

192.168.0.0/24

256 IPs

Preview

VPC

ap-south-1a

Route tables (1)

Network connections (1)

Step 5: Create Virtual Private Network VPN-GW-AWS on AWS

Details

Name tag - *optional*
Creates a tag with a key of 'Name' and a value that you specify.

VPN-GW-AWS

Value must be 256 characters or less in length.

Autonomous System Number (ASN)

☒ Amazon default ASN

☐ Custom ASN

Tags

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs. Name tag helps you track your resources more easily. We recommend adding Name tag.

Key

Value - *optional*

Q Name X

Q VPN-GW-AWS X

Remove

Add new tag

You can add 49 more tags.

Step 6: Attach Virtual Private Network VPN-GW-AWS to Virtual Cloud VPC-AWS on AWS

Virtual private gateways (1/1) Info							Actions
Filter virtual private gateways							Create virtual private gateway
Virtual private gateway ID: vgw-0d685dc970cfebac4 X Clear filters							Attach to VPC
							Detach from VPC
							Manage tags
							Delete virtual private gateway
Name	Virtual private gateway ID	State	Type	VPC			
VPN-GW-AWS	vgw-0d685dc970cfebac4	Detached	ipsec.1	-	64512		

Attach to VPC Info

Details

Virtual private gateway ID

vgw-0d685dc970cfebac4

Available VPCs

Attach the virtual private gateway to this VPC.

vpc-0522b2d8b4b2935ec / VNET-AWS-vpc

Step 7: Create Customer Gateway Cust-GW-AWS on AWS

Virtual private gateways (1/1) [Info](#)

Filter virtual private gateways

Virtual private gateway ID: vgw-0d685dc970cfebac4 [X](#) [Clear filters](#)

Name	Virtual private gateway ID	State	Type	VPC	Amazon ASN
VPN-GW-AWS	vgw-0d685dc970cfebac4	Attached	ipsec.1	vpc-0522b2d8b4b2935ec VNE...	64512

Create customer gateway [Info](#)

A customer gateway is a resource that you create in AWS that represents the customer gateway device in your on-premises network.

Details

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

Value must be 256 characters or less in length.

BGP ASN [Info](#)
The ASN of your customer gateway device.

Value must be in 1 - 2147483647 range.

IP address [Info](#)
Specify the IP address for your customer gateway device's external interface.

Certificate ARN
The ARN of a private certificate provisioned in AWS Certificate Manager (ACM).

Device - optional
Enter a name for the customer gateway device.

Step 8: Enable Route Propagation in Route Table for VPN-GW-AWS on AWS

VPC > Route tables > rtb-061fc40c64ea35f55 > Edit route propagation

Edit route propagation

Route table basic details

Route table ID

 rtb-061fc40c64ea35f55

Edit route propagation

Virtual Private Gateway	Propagation
vgw-0d685dc970cfebac4 / VPN-GW-AWS	☒ Enable

Step 9: Create S2S VPN Connection AWS-conn-AZURE on AWS by providing Public IP of Azure VPN Gateway VPN-GW-US

Create VPN connection [Info](#)

Select the resources and additional configuration options that you want to use for the site-to-site VPN connection.

Details

Name tag - *optional*

Creates a tag with a key of 'Name' and a value that you specify.

Value must be 256 characters or less in length.

Target gateway type [Info](#)

- ☒ Virtual private gateway
- ☐ Transit gateway
- ☐ Not associated

Virtual private gateway

Customer gateway [Info](#)

- ☒ Existing
- ☐ New

Customer gateway ID

Routing options [Info](#)

- ☐ Dynamic (requires BGP)
- ☒ Static

Static IP prefixes [Info](#)

Local IPv4 network CIDR - *optional*

The IPv4 CIDR range on the customer gateway (on-premises) side that is allowed to communicate over the VPN tunnels. The default is 0.0.0.0/0.

Remote IPv4 network CIDR - *optional*

The IPv4 CIDR range on the AWS side that is allowed to communicate over the VPN tunnels. The default is 0.0.0.0/0.

Step 10: Check the state of VPN Connection as 'Available' and then Download the Configuration from top right corner.

VPN connections (1/1) Info

Filter VPN connections

VPN ID: vpn-09a9288bd8a86b9cd Clear filters

Name	VPN ID	State	Virtual private gateway	Transit gateway	Customer gateway	Customer gateway ad...	Inside IP version	Type
AWS-conn-AZURE	vpn-09a9288bd8a86b9cd	Available	vgw-0d685dc970cfebac4	-	cgw-051fd337fde0a08ba	20.121.110.86	IPv4	ipsec.1

Download configuration

Choose the sample configuration you wish to download based on your customer gateway. Please note these are samples, and will need modification to use Advanced Algorithms, Certificates, and/or IPv6.

Vendor
The manufacturer of the customer gateway device (for example, Cisco Systems, Inc).

Generic

Platform
The class of the customer gateway device (for example, J-Series).

Generic

Software
The operating system running on the customer gateway device (for example, ScreenOS).

Vendor Agnostic

IKE version
The IKE version you are using for your VPN connection.

ikev2

Cancel Download

Step 11: From the configuration file copy the Pre-Shared Key which is to be used while creating Connection on Azure and Public IP of Virtual Private Gateway VPN-GW-AWS which is to be used in Local Network Gateway on Azure

```
IPSec Tunnel #1
=====
#1: Internet Key Exchange Configuration

Configure the IKE SA as follows:
Please note, these sample configurations are for the minimum requirement of AES128, SHA1, and DH Group 2.
Category "VPN" connections in the GovCloud region have a minimum requirement of AES128, SHA2, and DH Group 14.
You will need to modify these sample configuration files to take advantage of AES256, SHA256, or other DH groups like 2, 14-18,
22, 23, and 24.
NOTE: If you customized tunnel options when creating or modifying your VPN connection, you may need to modify these sample
configurations to match the custom settings for your tunnels.

Higher parameters are only available for VPNs of category "VPN," and not for "VPN-Classic".
The address of the external interface for your customer gateway must be a static address.
Your customer gateway may reside behind a device performing network address translation (NAT).
To ensure that NAT traversal (NAT-T) can function, you must adjust your firewall rules to unblock UDP port 4500.
If not behind NAT, and you are not using an Accelerated VPN, we recommend disabling NAT-T. If you are using an Accelerated VPN,
make sure that NAT-T is enabled.
- IKE version      : IKEv2
- Authentication Method : Pre-Shared Key
- Pre-Shared Key    : RCBh8bl_lGpaQDoTcgKDryw2iFvN013y
- Authentication Algorithm : sha1
- Encryption Algorithm : aes-128-cbc
- Lifetime          : 28800 seconds
- Phase 1 Negotiation Mode : main
- Diffie-Hellman     : Group 2

The Customer Gateway inside IP address should be configured on your tunnel
interface.

Outside IP Addresses:
- Customer Gateway      : 20.121.110.86
- Virtual Private Gateway : 15.207.187.183

Inside IP Addresses
- Customer Gateway      : 169.254.22.122/30
- Virtual Private Gateway : 169.254.22.121/30
```

Step 12: Create Local Network Gateway Local-GW-US on Azure and fill in the Public IP of Virtual Private Gateway VPN-GW-AWS copied from downloaded configuration file from AWS.

Microsoft Azure

Search resources, services, and docs (G+J)

Home > Local network gateways >

Create local network gateway ...

Basics Advanced Review + create

A local network gateway is a specific object that represents an on-premises location (the site) for routing purposes. [Learn more](#)

Project details

Subscription * Azure subscription 1

Resource group * RG-US [Create new](#)

Instance details

Region * East US

Name * Local_GW-US ✓

Endpoint ⓘ IP address FQDN

IP address * ⓘ 15.207.187.183 ✓

Address Space(s) ⓘ

192.168.0.0/16 ✓ ...

Add additional address range

Step 13: Create Connection Azure-conn-AWS on Azure and provide the Pre-Shared Key copied from downloaded configuration file from AWS

The screenshot shows the 'Create connection' page in the Microsoft Azure portal. The 'Settings' tab is selected. The 'Virtual network gateway' section includes the following configuration:

- Virtual network gateway: VP-GW-US
- Local network gateway: Local_GW-US
- Shared key (PSK): [Redacted]
- IKE Protocol: IKEv2
- Use Azure Private IP Address: [Unchecked]
- Enable BGP: [Checked]
- IPsec / IKE policy: Default
- Use policy based traffic selector: Disable
- DPD timeout in seconds: 45
- Connection Mode: Default

The 'NAT Rules Associations' section shows 'Ingress NAT Rules' and 'Egress NAT Rules' both set to '0 selected'.

Step 14: When Connection is established between Azure and AWS the status will be shown 'Connected'

The screenshot shows the 'Connections' page in the Microsoft Azure portal. A single connection is listed:

Name	Status	Peer 1	Peer 2	Resource group	Location	Subscription
Azure-conn-AWS	Connected	vp-gw-us	local_gw-us	RG-US	East US	Azure subscription 1

Step 15: Tunnel Status of VPN Connection on AWS will be shown 'UP'

The screenshot shows the 'VPN connections' page in the AWS Management Console. The connection 'AWS-conn-AZURE' is shown with a status of 'Available'. Below the connection list, the 'Tunnel state' is detailed:

Tunnel number	Outside IP address	Inside IPv4 CIDR	Inside IPv6 CIDR	Status	Last status change	Details	Certificate ARN
Tunnel 1	15.207.187.183	169.254.22.120/30	-	Up	February 10, 2023, 22:09:55 (UTC+05:30)	-	-
Tunnel 2	35.154.30.38	169.254.252.248/30	-	Up	February 10, 2023, 22:29:42 (UTC+05:30)	-	-

Step 17: Launch VM Instance for Windows server under VNET-US subnet on Azure

Microsoft Azure

Search resources, services, and docs (G+)

[Home](#) > [Virtual machines](#) >

Create a virtual machine

Subscription * ⓘ

Azure subscription 1

Resource group * ⓘ

RG-US

[Create new](#)

Instance details

Virtual machine name * ⓘ

vm-US

Region * ⓘ

(US) East US

Availability options ⓘ

Availability zone

Availability zone * ⓘ

Zones 1

You can now select multiple zones. Selecting multiple zones will create one VM per zone. [Learn more](#)

Security type ⓘ

Standard

Image * ⓘ

Windows Server 2019 Datacenter - x64 Gen2

[See all images](#) | [Configure VM generation](#)

VM architecture ⓘ

☐ Arm64

☒ x64

Arm64 is not supported with the selected image.

Run with Azure Spot discount ⓘ

☐

Size * ⓘ

Standard_DS1_v2 - 1 vcpu, 3.5 GiB memory (₹7,223.60/month)

[See all sizes](#)

Administrator account

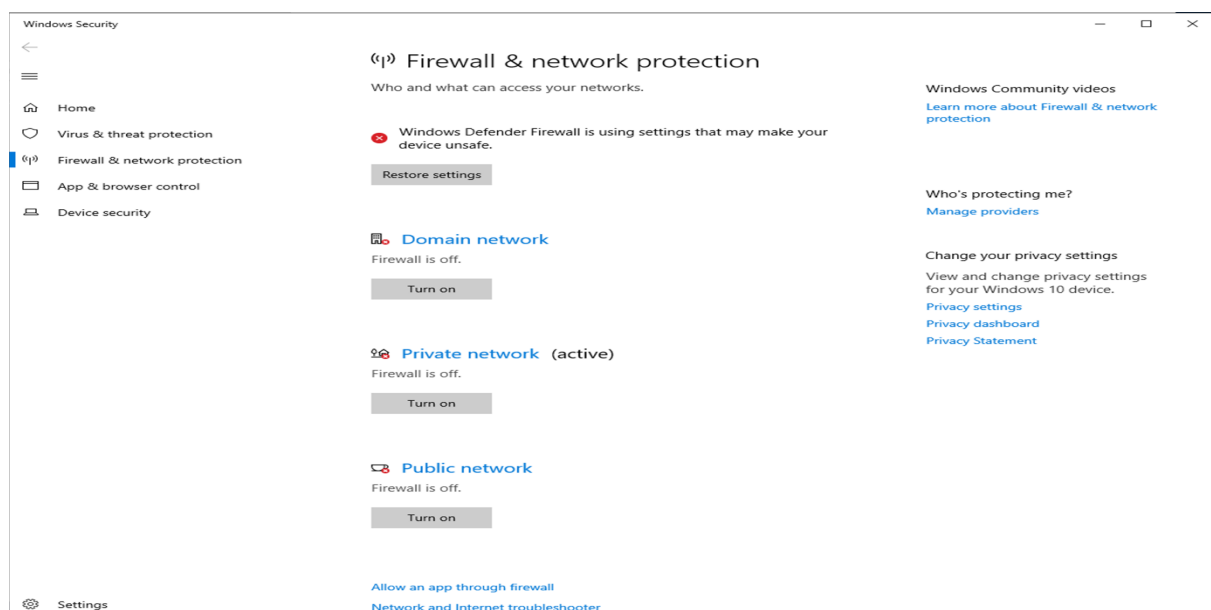
Username * ⓘ

Administrator123

Password * ⓘ

Confirm password * ⓘ

Step 18: Turn off Firewall on both the EC2 and VM instances on AWS and Azure after taking RDP



Conclusion:

- Azure instance is able to ping AWS instance from IP 10.0.0.4 to 192.168.0.171

```
Administrator: C:\Windows\system32\cmd.exe

C:\Users\Administrator123>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : 5wtzcgjhjje5bqzhd4uymle.bx.internal.cloudapp.net
    Link-local IPv6 Address . . . . . : fe80::4f56:2f2a:9e1b:9406%6
    IPv4 Address. . . . . : 10.0.0.4
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.0.1

C:\Users\Administrator123>ping 192.168.0.171

Pinging 192.168.0.171 with 32 bytes of data:
Reply from 192.168.0.171: bytes=32 time=209ms TTL=128
Reply from 192.168.0.171: bytes=32 time=209ms TTL=128
Reply from 192.168.0.171: bytes=32 time=210ms TTL=128
Reply from 192.168.0.171: bytes=32 time=209ms TTL=128

Ping statistics for 192.168.0.171:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 209ms, Maximum = 210ms, Average = 209ms

C:\Users\Administrator123>_
```

- AWS instance is able to ping Azure instance from IP 192.168.0.171 to 10.0.0.4

```
Administrator: C:\Windows\system32\cmd.exe

C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : ap-south-1.compute.internal
    Link-local IPv6 Address . . . . . : fe80::8ac4:ff7:2c55:ba0b%4
    IPv4 Address. . . . . : 192.168.0.171
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1

C:\Users\Administrator>ping 10.0.0.4

Pinging 10.0.0.4 with 32 bytes of data:
Reply from 10.0.0.4: bytes=32 time=209ms TTL=128
Reply from 10.0.0.4: bytes=32 time=213ms TTL=128
Reply from 10.0.0.4: bytes=32 time=209ms TTL=128
Reply from 10.0.0.4: bytes=32 time=210ms TTL=128

Ping statistics for 10.0.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 209ms, Maximum = 213ms, Average = 210ms

C:\Users\Administrator>_
```